

JUNGMIN KWON

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RESEARCH INTERESTS

Federated Learning, Network Coding, Data Imputation, Edge Intelligence, Machine Learning-Driven Networked Systems, Wireless Ad-Hoc Networks, Abnormal Network Data Detection

ON-GOING RESEARCH

Edge Intelligence & Federated Learning

- Researching approximated model parameter transmission strategies for federated learning
- *Keywords*: federated learning, low-rank approximation, unreliable network, systematic network coding
- Researching efficient model distribution methods in decentralized federated learning systems
- *Keywords*: federated learning, network coding, network topology analysis
- Analyze network data and research intelligent control systems in RAN environments
- *Keywords*: network data, open RAN, machine learning

ON-GOING PROJECT

- Development of Artificial Intelligent Control Technologies in xHaul Networks for 6G Open RAN Co-Design and Virtual Simulation-Based Verification
- Supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. RS-2025-00556192, Sejong Science Fellowship Domestic Track)
- Effective start/end date: 1/3/2025 - 28/2/2030
- Funding (KRW): ₩600,000,000
- Integrated ADMET/PK Prediction Platform Powered by Agentic AI and Federated Learning
- the Ministry of Health & Welfare and Ministry of Science and ICT, Republic of Korea (No. RS-2026-25607354)
- Effective start/end date: 7/1/2026 - 12/31/2028
- Funding (KRW): ₩250,000,000

EXPERIENCE

Kangwon National University

Assistant Professor

Sep. 2025 - Present

- Department of Electrical and Electronics Engineering
- Fields: Federated Learning, Low-Rank Model Learning, Network Coding

Ewha Womans University

Postdoctoral Researcher

Sep. 2024 - Aug. 2025

- Department of Electronic and Electrical Engineering
- Fields: Federated Learning, Low-Rank Model Learning, Network Coding
- Advisor: Prof. Hyunggon Park

Ph.D.

Mar. 2017 - Aug. 2024

- Department of Electronic and Electrical Engineering
- Fields: Distributed Data Analytics, Network Science, Low-Rank Model Learning, Complex Network
- Advisor: Prof. Hyunggon Park
- Dissertation: Approximate Strategies for Data Recovery and Transmission in Unstable Communication Networks

M.A.	Mar. 2015 - Feb. 2017
- Department of Electronic and Electrical Engineering	
- Fields: Network Coding, Distributed Data Analytics	
- Advisor: Prof. Hyunggon Park	
- Dissertation: Matrix Decomposition based Low Complexity Algorithms for Network Coding	
- GPA: 3.96/4.0 (99.1%)	
B.A.	Mar. 2011 - Feb. 2015
- Department of Electronic and Electrical Engineering	
- GPA: 3.82/4.0 (97.6%, <i>Summa cum laude</i>)	

PUBLICATIONS

Journal Articles

1. Jonghoon Park*, Dongyong Choi*, Kabseok Ko, Kuduck Kwon, **Jungmin Kwon**, In-Chul Hwang, and Seungyoung Park, "Batch Bayesian Optimization for Minimizing IPN in RF PLLs," *Journal of Semiconductor Technology and Science*, vol. 26, no. 5, pp. 327-339, Oct. 2026.
2. **Jungmin Kwon** and Hyunggon Park, "Resilient and Efficient Approximate Data Recovery for Federated Learning," *IEEE Transactions on Mobile Computing*, vol. 25, no. 5, pp. 6413-6428, May 2026.
3. **Jungmin Kwon**, Sujin Ahn, Hyunggon Park and Minhae Kwon, "Stochastic Approximate Inference of Latent Information in Epidemic Model," *Signal Processing*, vol. 232, pp. 109919, Jul. 2025.
4. **Jungmin Kwon** and Hyunggon Park, "Parameter Transferred Irreducible LSTM for Traffic Data Imputation," *IEEE Sensors Journal*, vol. 24, no. 14, pp. 22178-22188, Jul. 2024.
5. **Jungmin Kwon** and Hyunggon Park, "Data Dissemination Framework Using Low-Rank Approximation in Edge Networks," *IEEE Access*, vol. 12, pp. 1266-1279, Jan. 2024.

International Conference

1. Suhyun Han and **Jungmin Kwon**, "Contribution-Aware LoRA Rank Allocation for Federated LLM Fine-Tuning," International Conference on ICT Convergence (ICTC 2026), Oct. 2026.
2. Dahae Lee and **Jungmin Kwon**, "Integrated Detection and Latency-Aware XGBoost Intrusion Detection for ROS2," International Conference on ICT Convergence (ICTC 2026), Oct. 2026.
3. Dayoung Choi, Siyoun Park, **Jungmin Kwon**, and Hyunggon Park, "Few Features are Enough: Communication-Efficient AI-RAN," Conference on Neural Information Processing Systems (NeurIPS 2025) (AI and ML for Next-Generation Wireless Communications and Networking (AI4NextG)), Dec. 2025.
4. Dayoung Choi, Siyoun Park, **Jungmin Kwon**, and Hyunggon Park, "Interpretable KPI Analytics for Resource-Efficient AI-RAN Intelligence," Conference on Neural Information Processing Systems (NeurIPS 2025) (20th Women in Machine Learning Workshop (WiML 2025)), Dec. 2025.
5. **Jungmin Kwon** and Hyunggon Park, "Comparative Analysis of Network Coding Algorithms in Centralized Federated Learning over Unreliable Networks," International Conference on Ubiquitous and Future Networks (ICUFN 2025), Jul. 2025.
6. **Jungmin Kwon**, and Hyunggon Park, "Missing Traffic Data Imputation Using Multi-Trajectory Parameter Transferred LSTM," Conference on Neural Information Processing Systems (NeurIPS 2022) (Machine Learning for Autonomous Driving Workshop), Dec. 2022.
7. **Jungmin Kwon**, and Hyunggon Park, "Efficient Low-rank Federated Learning based on Singular Value Decomposition," The ACM International Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc 2022) Oct. 2022, pp.285-286.
8. **Jungmin Kwon**, Chaeyeon Cha, and Hyunggon Park, "Multilayered LSTM with Parameter Transfer for Vehicle Speed Data Imputation," IEEE International Symposium on Circuits and Systems (IEEE ISCAS 2021), May. 2021.

9. **Jungmin Kwon**, Chaeyeon Cha, and Hyunggon Park, “Bidirectional Imputation of Spatio-Temporal Data based on LSTM with Parameter Transfer,” IEEE Wireless Communications and Networking Conference (IEEE WCNC 2021), Apr. 2021.
10. **Jungmin Kwon**, Chaeyeon Cha, and Hyunggon Park, “Vehicle Speed Data Imputation based on Parameter Transferred LSTM,” Conference on Neural Information Processing Systems (NeurIPS 2020) (Machine Learning for Autonomous Driving Workshop), Dec. 2020.
11. **Jungmin Kwon**, Jungjin Lee, Miseon Yu, and Hyunggon Park, “Automatic Classification of Network Traffic Data based on Deep Learning in ONOS Platform,” International Conference on ICT Convergence (ICTC 2020), Oct. 2020, pp. 1041-1043.
12. **Jungmin Kwon**, Daeun Jung, and Hyunggon Park, “Traffic Data Classification using Machine Learning Algorithms in SDN Networks,” International Conference on ICT Convergence (ICTC 2020), Oct. 2020, pp. 1044-1046.
13. **Jungmin Kwon** and Hyunggon Park, “Efficient and Reliable Data Dissemination over Handover Dynamics in V2I Networks,” IEEE International Conference on Consumer Electronics (IEEE ICCE 2020), Jan. 2020.
14. **Jungmin Kwon** and Hyunggon Park, “Reliable Data Dissemination Strategy based on Systematic Network Coding in V2I Networks,” International Conference on ICT Convergence (ICTC 2019), Oct. 2019.
15. **Jungmin Kwon**, Sunwoo Cho, Sujin Kim, Yoonji Park, and Hyunggon Park, “Implementation of Network Coding Algorithm for Single Queue with Single Memory Devices,” International Conference on Ubiquitous and Future Networks (ICUFN 2017), Jul. 2017.
16. Minhae Kwon, **Jungmin Kwon**, Byungchul Park, and Hyunggon Park, “An Architecture of IPTV Networks Based on Network Coding,” International Conference on Ubiquitous and Future Networks (ICUFN 2017), Jul. 2017.
17. **Jungmin Kwon** and Hyunggon Park, “Low Complexity Algorithms for Network Coding Based on Singular Value Decomposition,” International Conference on Ubiquitous and Future Networks (ICUFN 2016), Jul. 2016, pp. 641-643
18. **Jungmin Kwon**, Minhae Kwon, and Hyunggon Park, “Energy-Efficient Data Transmission Strategy for Network Coded Bidirectional De Bruijn Networks,” International Conference on Ubiquitous and Future Networks (ICUFN 2015), Jul. 2015, pp. 37-39.
19. **Jungmin Kwon**, Minhae Kwon, Bungchul Park, and Hyunggon Park. “Implementation of Approximate Decoding Algorithm Over TCP/IP,” International Conference on Ubiquitous and Future Networks (ICUFN 2014), Jul. 2014. pp. 475-476.

PATENTS

International Registered Patents

1. **Jungmin Kwon**, Hyunggon Park, “Method and Device for Data Transmission in V2I Network,” US Patent, 2020.8.25 (Application no. 17/002,301), 2022.06.14 (Patent No. US 11363432).
2. **Jungmin Kwon**, Hyunggon Park, “Method and Apparatus for Network Encoding based on Low-Complexity Algorithms for Network Coding using Matrix Decomposition,” US Patent, 2018.04.19 (Application no. US 15/957689), 2020.12.01 (Patent No. US 10855817)
3. **Jungmin Kwon**, Hyunggon Park, Minhae Kwon, Soonyoung Kim, Minji Lee, Yeseul Jo, “Method For Propagating Network Management Data for Energy-Efficient IoT Network Management and Energy-Efficient IoT Node Apparatus,” US Patent, 2015.06.24 (Application no. US 14/748334), 2017.10.17 (Patent No. US 9794122)

International Pending Patents

1. **Jungmin Kwon**, Chaeyeon Cha, Hyunggon Park, “Lost Data Recovery Method and Apparatus using Parameter Transfer LSTM,” PCT, 2021. 10. 21 (Application no. PCT/KR2021/014824)
2. **Jungmin Kwon**, Hyunggon Park, “Reliable Data Dissemination Strategy based on Systematic Network Coding in V2I Networks,” US provisional Application, 2019.11.25 (Application no. 62/939,752)

3. **Jungmin Kwon**, Hyunggon Park, “Efficient and Reliable Data Dissemination based on Systematic Network Coding in V2I Networks,” US provisional Application, 2019.11.25 (Application no. 62/939,750)
4. **Jungmin Kwon**, Minhae Kwon, Hyunggon Park, “Sensor Node and Data Transmission Method of Sensor Node, Sink Node and Data Transmission Method of Sink Node,” PCT, Mar. 13, 2015 (Application no. PCT/KR2015/002434)

Domestic Registered Patents

1. **Jungmin Kwon**, Chaeyeon Cha, Hyunggon Park, Nayoung Kim, Daeun Jung, “Method and Apparatus for Predicting Demand for Public Goods Using a Machine Learning Model,” Korean Patent, filed on Dec. 31, 2021 (Application No. 10-2021-0193835), issued on Nov. 18, 2024 (Patent No. 10-2732984)
2. **Jungmin Kwon**, Chaeyeon Cha, Hyunggon Park, “Lost Data Recovery Method and Apparatus using Parameter Transfer LSTM,” Korean Patent, filed on Nov. 13, 2020 (Application No. 10-2020-0152145), issued on Mar. 17, 2022 (Patent No. 10-2377474)
3. **Jungmin Kwon** and Hyunggon Park, “Efficient and reliable data dissemination system between handovers in V2I networks,” Korean Patent, filed on Nov. 26, 2019 (Application No. 10-2019-0153406), issued on Apr. 2, 2021 (Patent No. 10-2238018)
4. **Jungmin Kwon** and Hyunggon Park, “High fidelity data dissemination method based on the predicted amount of data information in V2I networks,” filed on Nov. 26, 2019 (Application No. 10-2019-0153440), issued on Mar. 3, 2021 (Patent No. 10-2226794)
5. **Jungmin Kwon** and Hyunggon Park, “Apparatus and Method for Network Encoding based on Low Complexity Algorithms for Network Coding using Matrix Decomposition,” Korean Patent, filed on May 12, 2017 (Application no. 10-2017-0059396), issued on Dec. 26, 2018 (Patent no. 10-1934638)
6. **Jungmin Kwon** and Hyunggon Park, “Apparatus and Method for Network Encoding and Decoding based on Low Complexity algorithms for Network Coding using Singular Value Decomposition,” Korean Patent, Jun. 22, 2017 (Application no. 10-2017-0079210), issued on Jul. 2, 2018 (Patent no. 10-1876130)
7. **Jungmin Kwon**, Hyunggon Park, Yunji Park, Yongjin Jeong, Seongsung Yoon, and Sunwoo Cho, “Apparatus and Method for Network Encoding based on Low Complexity Algorithms for Network Coding using Unitary Hessenberg Matrix,” Korean Patent, filed on Jun. 22, 2017 (Application no. 10-2017-0079244), issued on Jul. 2, 2018 (Patent no. 10-1876131)
8. **Jungmin Kwon**, Hyunggon Park, Minhae Kwon, Sanghee Park, Jungmin Park, Minju Kim, “Apparatus and Method for Encoding or Decoding a Transmission Packet Including Parity,” Korean Patent, filed on Mar. 7, 2016 (Application no. 10-2016-0026924), issued on Nov. 1, 2017 (Patent no. 10-1795217)
9. **Jungmin Kwon**, Hyunggon Park, Minhae Kwon, Hyunji Na, Yujeong Lee, “Apparatus and Method for Encoding or Decoding a Message,” Korean Patent, filed on Mar. 7, 2016 (Application no. 10-2016-0026907), issued on Nov. 1, 2017 (Patent no. 10-1795216)
10. **Jungmin Kwon**, Hyunggon Park, and Minhae Kwon “System for High Resolution Live Broadcasting based on Network Coding,” Korean Patent, filed on Jun. 23, 2015 (Application no. 10-2015-0089363), issued on Jul. 22, 2016 (Patent no. 10-1643848)
11. **Jungmin Kwon**, Hyunggon Park, Byungchul Park, Minhae Kwon, “TCP based Network Routers Supporting Network Coding and TCP based Data Streaming System using Network Coding,” Korean Patent, Jan. 9, 2015 (Application no. 10-2015-0003653), issued on Jan. 19, 2016 (Patent no. 10-1588549)
12. **Jungmin Kwon**, Hyunggon Park, and Minhae Kwon “Sensor Node and Data Transmission Method of Sensor Node, Sink Node and Data Transmission Method of Sink Node,” Korean Patent, filed on Mar. 14, 2014 (Application no. 10-2014-0030594), issued on Oct. 28, 2015 (Patent no. 10-1565685)
13. **Jungmin Kwon**, Hyunggon Park, Minhae Kwon, Soonyoung Kim, Mini Lee, and Yeseul Cho, “Method for Propagating Network Management Data for Energy-Efficient IoT Network Management and Energy-Efficient IoT Node Apparatus,” Korean Patent, filed on Jun. 24, 2014 (Application no. 10-2014-0077225), issued on Sep. 17, 2015 (Patent no. 10-1555315)

RESEARCH EXPERIENCES

Visiting Ph.D. Student

- Swiss Federal Institute of Technology Lausanne (EPFL), Lausanne, Switzerland (Jun. 2018 - Aug. 2018)
 - Signal Processing Laboratory LTS4, Electrical Engineering
 - Advisor: Prof. Pascal Frossard
 - Supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (Young Researchers' Exchange Programme between Korea and Switzerland 2017-2018)
 - Funding (CHF): ₩2,500
- University of Cambridge, Cambridge, United Kingdom (Jan. 2020 - Mar. 2020)
 - Machine Learning and Artificial Intelligence for Medicine Group, Electrical and Computer Engineering Department
 - Advisor: Prof. Mihaela van der Schaar

HONORS AND AWARDS

- Best Paper Award, KICS, Nov. 2015, Nov. 2024.
- Merit based scholarship (granted for full semesters), The Samil Foundation, 2018
- Qualcomm Innovation Awards, Qualcomm Inc., June 1, 2017
- Merit based scholarship (granted for full semesters), Ewha Womans University, 2017
- Received an encouragement award in the Research Proposal Contest using Sex and Gender Analysis Methodology, WISSET, Nov.11, 2015
- TOP10 competent person scholarship (granted for full semesters), Ewha Womans University, (2015 & 2016)
- Kim Aeda Award, given to the best GPA (summa cum laude) in College of Engineering, Ewha Womans University, Feb.23, 2015
- WISSET-KICS Young Female Researcher Award, WISSET & KICS, Nov.22, 2014
- Merit based scholarship (granted for full semesters), Kwanjeong Educational Foundation (2013 & 2014)

SKILLS

Languages: English(fluent), Korean(native)

Programming: Python, C/C++, LaTeX

ML/DL Frameworks: PyTorch, TensorFlow, Matlab

OS Environments: Linux, Mac, Windows, Anaconda, Docker